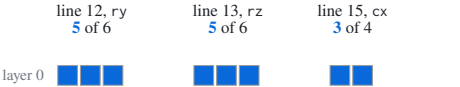
① srcML of the loop nest (lines 10–15)

```
<for>e for <control><expr><name>layer</name></expr><range>in <expr><call><name>range</name><argument_list>(<argument>
    ↪ <expr><name>layers[f]</name></expr></argument>)</argument_list></call></expr></range></control><block>:<block_content>
    <for>e for <control><expr><name>qubit</name></expr><range>in <expr><call><name>range</name><argument_list>(
        ↪ <argument><expr><name>n[f]</name></expr></argument>)</argument_list></call></expr></range></control><block>:
        ↪ <block_content>
        <expr_stmt><expr><call><name><name>qc</name><operator>.</operator><name>ry[b]</name></name><argument_list>(
            ↪ <argument>angle…</argument>, <argument><expr><name>qubit[c]</name></expr></argument>)</argument_list></call></expr>
            ↪ </expr_stmt>
        <expr_stmt><expr><call><name><name>qc</name><operator>.</operator><name>rz[b]</name></name><argument_list>(
            ↪ <argument>angle…</argument>, <argument><expr><name>qubit[c]</name></expr></argument>)</argument_list></call></expr>
            ↪ </expr_stmt>
</block_content></block></for><for>e for <control><expr><name>qubit</name></expr><range>in <expr><call><name>
    ↪ range</name><argument_list>(<argument><expr><name>n[f]</name><operator>-</operator><literal type="number">1</literal>
    ↪ </expr></argument>)</argument_list></call></expr></range></control><block>:<block_content>
        <expr_stmt><expr><call><name><name>qc</name><operator>.</operator><name>cx[b]</name></name><argument_list>(
            ↪ <argument><expr><name>qubit[c]</name></expr></argument>, <argument><expr><name>qubit[c]</name><operator>+
            ↪ </operator><literal type="number">1[c]</literal></expr></argument>)</argument_list></call></expr></expr_stmt>
</block_content></block></for></block_content></block></for>
```

e `for` → unrolled, iteration numbers kept **f** range → iteration count, from lines 4–5 (***n***=3, *layers*=2)
b gate name **c** `qubit`, `qubit + 1` → a qubit per iteration *angle…*: argument elided; the tool keeps it symbolic

② Unrolled operations in the slice of q_0



Filled: in the wire-rule slice of q_0 (13 of 16 operations). The lines are the same in every iteration, but in the second layer the rotations of q_2 and its `cx` *are not dependences of q_0; a line-level slice keeps lines 12, 13 and 15 anyway.*